

Name: **Ananya Sharma** Roll No: **27** Class: **X - B****Ans 1.**

Newton's second law states that the rate of change of momentum of a body is directly proportional to the applied force, and takes place in the direction of the force. $F = ma$.

Ans 2.

Speed is a scalar quantity, it only has magnitude, e.g. 40 km/hr. Velocity is a vector quantity, it has magnitude and direction, e.g. 40 km/hr towards north.

Ans 4.

SI unit of electric current is the Ampere (A). One ampere is the current when one coulomb of charge flows through a conductor in one second.

$$\text{Given } u = 0, a = 2 \text{ m/s}^2, t = 5 \text{ s.}$$

$$v = u + at = 0 + 2 \times 5 = 10 \text{ m/s.}$$

Ans 6. Distance, $s = ut + \frac{1}{2}at^2 = 0 + \frac{1}{2}(2)(25) = 25 \text{ m.}$ **Ans 8.**

Ohm's law: $V = IR$, at constant temperature current is proportional to voltage.

$$V = I \times R = 2 \times 10 = 20 \text{ Volts.}$$

$$\text{We know } v = u + at \dots (1)$$

$$\text{Also, average velocity} = \frac{(u+v)}{2}$$

$$\text{distance } s = \text{average velocity} \times \text{time}$$

$$s = \left[\frac{(u+v)}{2} \right] \times t$$

$$\text{substituting } v \text{ from (1):}$$

Ans 11 (a). $s = \left[\frac{(u + u + at)}{2} \right] \times t = ut + \frac{1}{2}at^2$

(answer continues on next page...)

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Hence proved, $s = ut + \frac{1}{2}at^2$ is the second equation of motion for uniformly accelerated linear motion.

Ans 9.

The three states of matter are solid, liquid and gas. In solids particles are tightly packed with strong forces. In liquids particles can move around but stay close. In gases particles are far apart and move freely.

Ans 7.

Mitosis is cell division producing two identical daughter cells, used for growth and repair. Meiosis produces four daughter cells with half the chromosome number, used for reproduction (gametes).

Ans 12.

Two ways to conserve water: (1) Rainwater harvesting - collecting rain water for reuse. (2) Repairing leaking pipes and taps promptly to avoid wastage.

Physics formula sheet - just for my own revision:

$F = ma$, $v = u + at$, $s = ut + \frac{1}{2}at^2$, $v^2 = u^2 + 2as$,

Ohm's law $V=IR$, Power $P = VI$